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JEE MAIN PHYSICS DPP: MOMENTUM & NEWTON'S 2ND LAW

Topic: Linear Momentum, Newton's 2nd Law of Motion & Variable Mass Systems | Time: 45 Min | Max Marks: 80

Marking Scheme: **+4, -1** | Designed & Curated by: **Syed Mazhar Ali**

Q1. The linear momentum of a particle of mass $m = 2$ kg varies with time as $\vec{p}(t) = (3t^2\hat{i} + 4t\hat{j})$ kg · m/s. The magnitude of the net force acting on the particle at $t = 2$ s is:

(a) 12 N
(b) $4\sqrt{10}$ N ≈ 12.65 N
(c) 16 N
(d) 8 N

Q2. A force $\vec{F} = (6t^2\hat{i} + 4t\hat{j})$ N acts on a body of mass 3 kg initially at rest at the origin. The velocity of the body at $t = 3$ s is:

(a) $(18\hat{i} + 9\hat{j})$ m/s
(b) $(54\hat{i} + 18\hat{j})$ m/s
(c) $(18\hat{i} + 6\hat{j})$ m/s
(d) $(6\hat{i} + 6\hat{j})$ m/s

Q3. Water from a hose pipe of cross-sectional area $A = 10$ cm^2 strikes a vertical wall normally with a speed of $v = 15$ m/s and falls dead down the wall without bouncing. If the density of water is $\rho = 1000$ kg/m^3, the force exerted by the water stream on the wall is:

(a) 150 N
(b) 450 N
(c) 300 N
(d) 225 N

Q4. In the previous question, if the water stream bounces back elastically with the same speed from the wall, the force exerted on the wall would be:

(a) 450 N
(b) 225 N
(c) 900 N
(d) 112.5 N

Q5. Sand is dropped onto a conveyor belt at a constant rate of $\frac{dm}{dt} = 2$ kg/s. To keep the belt moving at a constant speed of $v = 3$ m/s, the extra force required to drive the belt is:

(a) 3 N
(b) 6 N
(c) 12 N
(d) 1.5 N

Q6. In Q5, the extra power required to keep the conveyor belt moving at constant speed $v = 3$ m/s is:

(a) 9 W
(b) 36 W
(c) 18 W
(d) 6 W

Q7. A rocket of initial mass $M_0 = 1000$ kg ejects gas at a constant exhaust speed $u = 500$ m/s relative to the rocket. To give the rocket an initial upward acceleration equal to $g = 10$ m/s^2, the rate of consumption of fuel ($\frac{dm}{dt}$) must be:

(a) 20 kg/s
(b) 10 kg/s
(c) 80 kg/s
(d) 40 kg/s

Q8. A machine gun fires $n = 360$ bullets per minute, each of mass $m = 20$ g, with a muzzle speed of $v = 500$ m/s. The average recoil force required to hold the gun steady in position is:

(a) 60 N
(b) 36 N
(c) 120 N
(d) 180 N

Q9. A particle of mass m moves along the x-axis such that its position is given by $x(t) = at^2 - bt^3$. The force acting on the particle is zero at time t equal to:

(a) $\frac{2a}{3b}$
(b) $\frac{a}{3b}$
(c) $\frac{a}{b}$
(d) $\frac{3a}{2b}$

Q10. Two bodies A and B have masses $m_A = 2$ kg and $m_B = 8$ kg and possess equal kinetic energies. The ratio of their linear momenta $\frac{p_A}{p_B}$ is:

(a) 1 : 4
(b) 4 : 1
(c) 1 : 2
(d) 2 : 1

Q11. A body of mass 5 kg starts from rest under the action of a force $\vec{F} = (10\hat{i} + 15\hat{j})$ N. The kinetic energy of the body at $t = 2$ s is:

(a) 65 J
(b) 260 J

(c) 520 J
(d) 130 J

Q12. A ball of mass $m = 0.2$ kg moving horizontally with velocity $u = 20\hat{i}$ m/s hits a smooth vertical wall and rebounds with velocity $v = -15\hat{i}$ m/s. If the impact lasts for $\Delta t = 0.01$ s, the average force exerted by the wall on the ball is:

(a) $-700\hat{i}$ N
(b) $+700\hat{i}$ N
(c) $-100\hat{i}$ N
(d) $-350\hat{i}$ N

Q13. The linear momentum p of a body increases by 50%. The percentage increase in its kinetic energy is:

(a) 50%
(b) 125%
(c) 100%
(d) 225%

Q14. A constant retarding force $F = 50$ N is applied to a body of mass 20 kg moving initially with speed 15 m/s. The time taken by the body to come to a complete stop is:

(a) 3.0 s
(b) 4.5 s
(c) 6.0 s
(d) 7.5 s

Q15. A body of mass m is moving in a circle of radius R with constant angular velocity ω. The rate of change of linear momentum over half a revolution is directed:

(a) Along the tangent to the path
(b) Outward from the center
(c) Zero
(d) Toward the center of the circle

Q16. An open rail wagon of mass $M_0 = 1000$ kg is coasting freely with velocity $v_0 = 10$ m/s along a straight horizontal track. Rain starts falling vertically into the wagon at a constant rate $\mu = 5$ kg/s. Neglecting friction, the velocity of the wagon after $t = 100$ s is:

(a) 6.67 m/s
(b) 5.00 m/s
(c) 7.50 m/s
(d) 8.33 m/s

Q17. A particle of mass m is subjected to a force $F(x) = -kx + \beta x^3$. The motion of the particle is:

(a) Simple Harmonic for all displacements
(b) Non-harmonic periodic for small oscillations near origin
(c) Uniformly accelerated
(d) Purely translational with constant momentum

Q18. A body of mass m is dropped from a height h. If air resistance exerts a retarding force proportional to speed ($F_{\text{drag}} = kv$), the terminal velocity of the body is:

(a) $\sqrt{2gh}$
(b) $\dfrac{k}{mg}$
(c) $\dfrac{mg}{k}$
(d) $\dfrac{m^2 g}{k}$

Q19. A block of mass $m = 1$ kg is moving on the x-axis under a conservative force. The potential energy is $U(x) = (x^2 - 4x)$ J. The magnitude of the force on the block at $x = 5$ m is:

(a) 10 N
(b) 4 N
(c) 5 N
(d) 6 N

Q20. A stone of mass $m = 0.5$ kg is tied to a string of length $L = 1$ m and rotated in a horizontal circle at constant speed $v = 4$ m/s on a frictionless table. The net force acting on the stone is:

(a) 8 N directed towards the center
(b) 8 N directed tangentially
(c) Zero
(d) 16 N directed radially outward

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	**6**	(c)	**11**	(d)	**16**	(a)
2	(c)	**7**	(d)	**12**	(a)	**17**	(b)
3	(d)	**8**	(a)	**13**	(b)	**18**	(c)
4	(a)	**9**	(b)	**14**	(c)	**19**	(d)
5	(b)	**10**	(c)	**15**	(d)	**20**	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

By Newton's second law:

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(3t^2\hat{i} + 4t\hat{j}) = 6t\hat{i} + 4\hat{j} \text{ N}$$

At $t = 2$ s:

$$\vec{F}(2) = 12\hat{i} + 4\hat{j} \text{ N}$$

$$|\vec{F}| = \sqrt{12^2 + 4^2} = \sqrt{160} = 4\sqrt{10} \text{ N}$$

Sol 2. Correct Option: (c)

Acceleration:

$$\vec{a}(t) = \frac{\vec{F}}{m} = \frac{6t^2\hat{i} + 4t\hat{j}}{3} = 2t^2\hat{i} + \frac{4}{3}t\hat{j} \text{ m/s}^2$$

Velocity at $t = 3$ s:

$$\vec{v}(3) = \int_0^3 \left(2t^2\hat{i} + \frac{4}{3}t\hat{j}\right) dt = \left[\frac{2}{3}t^3\hat{i} + \frac{2}{3}t^2\hat{j}\right]_0^3$$

$$\vec{v}(3) = \frac{2}{3}(27)\hat{i} + \frac{2}{3}(9)\hat{j} = 18\hat{i} + 6\hat{j} \text{ m/s}$$

Sol 3. Correct Option: (d)

Mass flow rate:

$$\frac{dm}{dt} = \rho A v = (1000)(10 \times 10^{-4})(15) = 15 \text{ kg/s}$$

Since water loses all its momentum upon hitting the wall:

$$F = \frac{dp}{dt} = v\frac{dm}{dt} = 15 \times 15 = 225 \text{ N}$$

Sol 4. Correct Option: (a)

For elastic reflection, change in velocity is $\Delta v = 2v$:

$$F = \Delta v\frac{dm}{dt} = 2v(\rho A v) = 2(225) = 450 \text{ N}$$

Sol 5. Correct Option: (b)

To accelerate the incoming stationary sand to belt speed v:

$$F_{\text{thrust}} = v_{\text{rel}}\frac{dm}{dt} = 3 \times 2 = 6 \text{ N}$$

Sol 6. Correct Option: (c)

Total power supplied by the motor:

$$P = F \times v = \left(v\frac{dm}{dt}\right)v = \frac{dm}{dt}v^2$$

$$P = 2 \times 3^2 = 18 \text{ W}$$

(Note: Rate of increase of kinetic energy is $\frac{1}{2}\frac{dm}{dt}v^2 = 9$ W, the remaining 9 W is dissipated as heat via friction).

Sol 7. Correct Option: (d)

Thrust force equation for upward acceleration a:

$$F_{\text{thrust}} - M_0 g = M_0 a$$

$$u\frac{dm}{dt} = M_0(g + a) = 1000(10 + 10) = 20000 \text{ N}$$

$$\frac{dm}{dt} = \frac{20000}{500} = 40 \text{ kg/s}$$

Sol 8. Correct Option: (a)

Firing rate: $N = \dfrac{360}{60} = 6$ bullets/s.
Mass of each bullet: $m = 20$ g $= 0.02$ kg.
Average recoil force:

$$F_{\text{avg}} = N\cdot(mv) = 6\times(0.02\times500) = 6\times10 = 60 \text{ N}$$

Sol 9. Correct Option: (b)

$v(t) = \dfrac{dx}{dt} = 2at - 3bt^2$.
$a(t) = \dfrac{dv}{dt} = 2a - 6bt$.
Force $F = ma(t) = 0 \implies 2a - 6bt = 0 \implies t = \dfrac{a}{3b}$.

Sol 10. Correct Option: (c)

Relation between momentum and kinetic energy:

$$p = \sqrt{2mK}$$

Since kinetic energies are equal:

$$\frac{p_A}{p_B} = \sqrt{\frac{m_A}{m_B}} = \sqrt{\frac{2}{8}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

Sol 11. Correct Option: (d)
$\vec{a} = \dfrac{\vec{F}}{m} = \dfrac{10\hat{i} + 15\hat{j}}{5} = 2\hat{i} + 3\hat{j}$ m/s^2.
Velocity at $t = 2$ s:

$$\vec{v}(2) = \vec{a}t = (2\hat{i} + 3\hat{j}) \times 2 = 4\hat{i} + 6\hat{j} \text{ m/s}$$

$v^2 = 4^2 + 6^2 = 16 + 36 = 52$ m^2/s^2.
Kinetic energy:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(5)(52) = 130 \text{ J}$$

Sol 12. Correct Option: (a)
Change in momentum of the ball:

$$\Delta\vec{p} = 0.2(-15\hat{i} - 20\hat{i}) = -7\hat{i} \text{ N} \cdot \text{s}$$

Average force:

$$\vec{F}_{\text{avg}} = \frac{\Delta\vec{p}}{\Delta t} = \frac{-7\hat{i}}{0.01} = -700\hat{i} \text{ N}$$

Sol 13. Correct Option: (b)
$K = \dfrac{p^2}{2m}$.
New momentum: $p' = 1.5p$.
New kinetic energy:

$$K' = \frac{(1.5p)^2}{2m} = 2.25K$$

Percentage increase:

$$\frac{K' - K}{K} \times 100 = (2.25 - 1) \times 100 = 125\%$$

Sol 14. Correct Option: (c)
Deceleration:

$$a = \frac{F}{m} = \frac{50}{20} = 2.5 \text{ m/s}^2$$

Time to stop:

$$t = \frac{u}{a} = \frac{15}{2.5} = 6.0 \text{ s}$$

Sol 15. Correct Option: (d)
By Newton's second law $\vec{F}_{\text{net}} = \dfrac{d\vec{p}}{dt}$. In UCM, the net force is the centripetal force directed strictly toward the center. Hence, $\dfrac{d\vec{p}}{dt}$ is directed towards the center.

Sol 16. Correct Option: (a)
Since rain falls vertically, no external horizontal force acts on the system ($F_{\text{ext},x} = 0$).
Conservation of horizontal momentum:

$$M_0 v_0 = (M_0 + \mu t)v(t)$$

$$(1000)(10) = (1000 + 5 \times 100)v(t)$$

$$10000 = 1500v(t) \implies v(t) = \frac{20}{3} \approx 6.67 \text{ m/s}$$

Sol 17. Correct Option: (b)
Near the origin, the cubic term βx^3 introduces non-linearity. The restoring force is non-linear ($F \propto -x +$ cubic terms), so the motion is periodic but non-harmonic.

Sol 18. Correct Option: (c)
At terminal velocity v_t, acceleration is zero ($a = 0$), so downward gravity balances upward drag:

$$mg - kv_t = 0 \implies v_t = \frac{mg}{k}$$

Sol 19. Correct Option: (d)
Force is the negative gradient of potential energy:

$$F(x) = -\frac{dU}{dx} = -(2x - 4) = 4 - 2x$$

At $x = 5$ m:

$$F(5) = 4 - 2(5) = -6 \text{ N} \implies |F| = 6 \text{ N}$$

Sol 20. Correct Option: (a)
Centripetal acceleration is $a_c = \dfrac{v^2}{L}$.
Net force (tension in string):

$$F_{\text{net}} = \frac{mv^2}{L} = \frac{0.5 \times 4^2}{1} = 8 \text{ N (towards center)}$$